

Project Code

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Research Question: “Which patient- or hospital-level factors are most predictive of readmission within 30 days among diabetic patients?”

Data Acquisition and Preprocessing

```
diabete <- read_csv("./diabetic_data_2.csv")
#head(diabete, 3)

# Handle missingness
diabete[diabete == "?"] <- NA

# Set missingness thresholds
col_na_thresh <- 0.05
row_na_thresh <- 0.05

# Filter out columns with >95% missing values
diabete <- diabete %>%
  dplyr::select(where(~ mean(is.na(.)) <= col_na_thresh))

# Filter out rows with >30% missing values
diabete <- diabete %>%
  filter(rowMeans(is.na(.)) <= row_na_thresh)

# Recode outcome variable
diabete <- diabete %>%
  mutate(readmission_binary = if_else(readmitted == "<30", 1, 0))
```

Exploratory Data Analysis

Model Development

Posterior Inference and Model evaluation

Interpretation and Reporting